

Cupcast I: A Dixon–Coles Model with Squad-Quality Shrinkage for Forecasting the 2026 FIFA World Cup

Carlos Garcia

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Abstract

I specify a forecasting model for the 2026 FIFA World Cup: a Dixon–Coles goal model [2] fitted by maximum likelihood on a decade of international results with exponential time-decay weighting, whose team ratings are shrunk toward a prior built from minutes-weighted, performance-based squad-quality composites. Match probabilities feed a Monte Carlo simulation of the full 48-team tournament under FIFA’s official group tiebreakers and Round-of-32 bracket allocation. This paper documents the data, the rating system, the model equations, the estimation procedure, and the simulation design. Validation is reported in Paper II; the forecast itself in Paper III.

1 Introduction

To be written as the pipeline lands.

2 Data

2.1 International match history

2.2 Squad and player data

2.3 Bookmaker odds

3 Elo Ratings

The rating input follows the World Football Elo system [6], whose predictive value for match outcomes is established empirically by Hvattum and Arntzen [3]. After each match the rating of team i updates as

$$R'_i = R_i + K G (W - W_e), \quad W_e = \frac{1}{1 + 10^{-(R_i - R_j + H)/400}}, \quad (1)$$

where K encodes match importance, G is the goal-difference multiplier, W the realised result, and H the home adjustment. Computing ratings from raw history (rather than scraping published tables) yields reproducible as-of-date ratings for backtesting.

4 The Goal Model

Goals scored by teams i and j are modelled as conditionally Poisson with rates

$$\lambda_i = \exp(\mu + \alpha_i - \beta_j + \gamma h_i), \quad \lambda_j = \exp(\mu + \alpha_j - \beta_i), \quad (2)$$

where α and β are attack and defense ratings, and the host indicator h_i is nonzero only for Mexico, the USA, and Canada in their home venues — the 2026 format makes every other fixture neutral. Independence is corrected at low scores by the Dixon–Coles adjustment [2]

$$\mathbb{P}(X = x, Y = y) = \tau_\rho(x, y) \frac{e^{-\lambda_i} \lambda_i^x}{x!} \frac{e^{-\lambda_j} \lambda_j^y}{y!}, \quad (3)$$

with τ_ρ reweighting the $\{0, 1\} \times \{0, 1\}$ scorelines; this addresses the draw underestimation of the independent-Poisson model of Maher [5]. Training matches are weighted by recency,

$$\phi(t) = \exp(-\xi \Delta t), \quad (4)$$

with ξ selected by out-of-sample predictive likelihood, following the evidence in Ley et al. [4] that smooth decay outperforms fixed windows for national teams. Friendlies receive an additional importance down-weight, mirroring the Elo K hierarchy.

5 Squad-Quality Composite

To be written.

6 Shrinkage Toward the Squad Prior

Fitted ratings are blended with the squad-composite-implied prior,

$$\tilde{\alpha}_i = w_i \hat{\alpha}_i + (1 - w_i) \alpha_i^{\text{prior}}, \quad w_i = \frac{n_i^{\text{eff}}}{n_i^{\text{eff}} + k}, \quad (5)$$

where n_i^{eff} is the decay-weighted match count, so teams with sparse competitive history lean on squad quality while data-rich teams keep their fitted ratings — the partial-pooling logic of hierarchical models [1] in a transparent two-stage form.

7 Tournament Simulation

To be written.

8 Reproducibility

To be written.

References

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